

4.3.6 Form and Div tag

i) HTML Form

HTML Form is a document which stores information of a user into a web server using interactive controls. An HTML form contains different kind of information such as user name, password, contact number, e-mail id, etc.

The elements used in an HTML form are check box, input box, radio buttons, submit buttons, etc. Using these elements, the information of the user is submitted into a web server.

Syntax:

```
<form action = "Script URL" method = "GET|POST">
```

form elements like input, textarea etc.

```
</form>
```

Attributes:

Attributes	Description
action	Backend script ready to process your passed data.
method	Method to be used to upload data. The most frequently used are GET and POST methods.
target	Specify the target window or frame where the result of the script will be displayed. It takes values like _blank, _self, _parent etc.
enctype	Specifies how the form-data should be encoded when submitting it to the server (only for method="post")

HTML Form Controls

There are different types of form controls that you can use to collect data using HTML form.

- Text input controls
- Check boxes controls
- Radio box controls
- Select box controls
- File select boxes
- Hidden controls
- Clickable buttons
- Submit and reset button

Text Input Controls

There are three types of text input used on forms:

- **Single-line text input control:** – This control is used for items that requires only one line of user input, such as search boxes or names. It is created using HTML `<input>` tag.

Attributes:

Attributes	Description
type	Indicates the type of input control and for text input control it will be set to text.
name	Used to give a name to the control which is sent to the server to be recognized and get the value.
value	This can be used to display default value inside the control.
size	Allows to specify the width of the text-input control in terms of characters.
maxlength	Allows to specify the maximum number of characters a user can enter into the text box.

Example:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Text Input Control</title>
```

```
</head>
```

```
<body>
```

```
<form >
```

```
First name: <input type="text" name="fname" value="Rudra Narayan ">
```

```
<br>
```

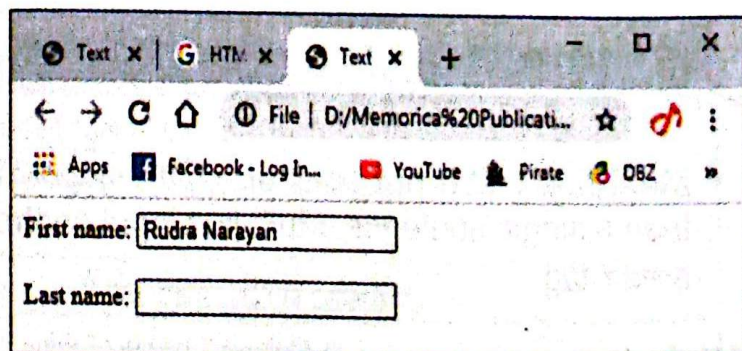
```
<br>
```

```
Last name: <input type="text" name="lname" >
```

```
</form>
```

```
</body>
```

```
</html>
```



- **Password input control:** – This is also a single-line text input but it masks the character as soon as a user enters it. They are also created using HTML `<input>` tag.

Example:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Password Input  
Control</title>
```

```
</head>
```

```
<body>
```

```
<form >
```

```
User ID : <input type = "text" name = "user_id" />
```

```
<br>
```

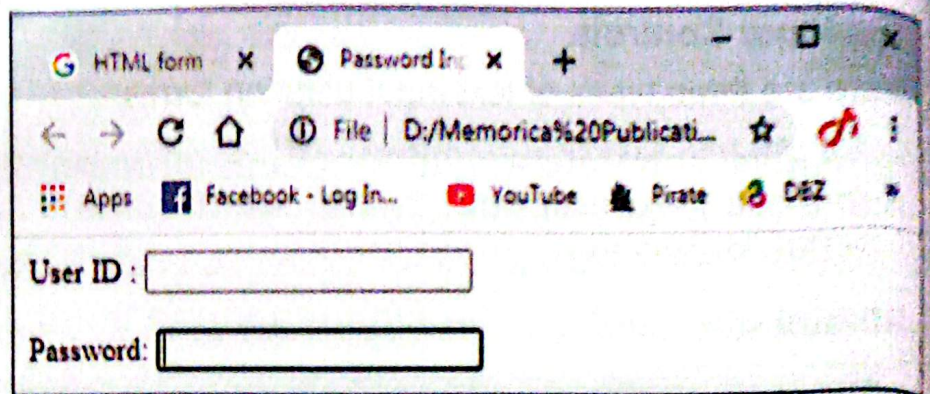
```
<br>
```

```
Password: <input type = "password" name = "password" />
```

```
</form>
```

```
</body>
```

```
</html>
```

A screenshot of a web browser window. The address bar shows a file path: D:/Memorica%20Publicati... The browser has several tabs open: 'HTML form' and 'Password In...'. The page content displays a form with two input fields. The first field is labeled 'User ID : ' and is a text input. The second field is labeled 'Password: ' and is a password input (masked with dots). The browser's toolbar includes back, forward, refresh, and home buttons, along with a search bar and several bookmarks like 'Apps', 'Facebook - Log In...', 'YouTube', 'Pirate', and 'DEZ'.

- **Multi-line text input control:** – This is used to give details that may be longer than a single sentence. Multi-line input controls are created using HTML `<text area>` tag.

Attributes:

Attributes	Description
rows	Indicates the number of rows of the text area box.
cols	Indicates the number of columns of the text area box.
name	Used to give a name to the control which is sent to the server to be recognized and get the value.

Example:

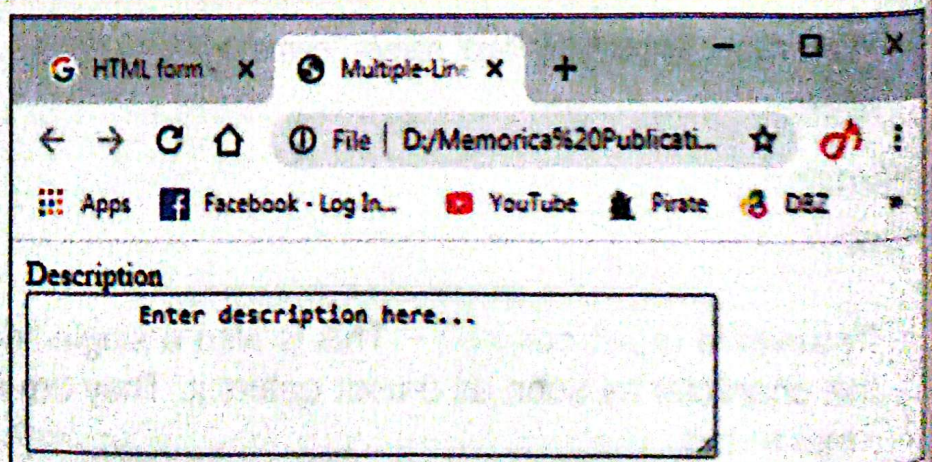
```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Multiple-Line Input  
Control</title>
```

```
</head>
```

A screenshot of a web browser window. The address bar shows a file path: D:/Memorica%20Publicati... The browser has several tabs open: 'HTML form' and 'Multiple-Line...'. The page content displays a form with a single multi-line text input field. Above the field is the label 'Description' and inside the field is the placeholder text 'Enter description here...'. The browser's toolbar includes back, forward, refresh, and home buttons, along with a search bar and several bookmarks like 'Apps', 'Facebook - Log In...', 'YouTube', 'Pirate', and 'DEZ'.

```

<body>
  <form>
    Description
    <br />
    <textarea rows = "5" cols = "50" name = "description">
      Enter description here...
    </textarea>
  </form>
</body>
</html>

```

Check box Control

Check boxes are used when more than one option need to be selected. They are also created using HTML <input> tag, but **type** attribute is set to check box.

Attributes:

Attributes	Description
type	Used to define the type of control button.
name	Used to give a name to the control which is sent to the server to be recognized and get the value.
value	The value that will be used if the check box is selected.
Checked	Set to 'checked' if you want to select it by default.

Example:

```

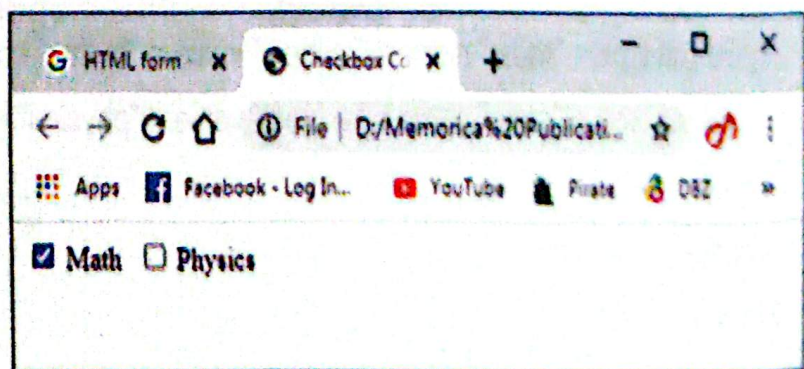
<!DOCTYPE html>
<html>
  <head>
    <title>Check box Control</title>
  </head>
  <body>
    <form>

```

```

      <input type = "checkbox" name = "math" value = "on" checked> Math
      <input type = "checkbox" name = "physics" value = "on"> Physics
    </form>
  </body>
</html>

```



```

    </form>
  </body>
</html>

```

Radio Button Control

Radio buttons are used when out of many options, just one option is required to be selected. They are also created using HTML `<input>` tag, but `'type'` attribute is set to `radio`.

Attributes:

Attributes	Description
type	Used to define the type of control button.
name	Used to give a name to the control which is sent to the server to be recognized and get the value.
value	The value that will be used if the radio box is selected.
Checked	Set to 'checked' if you want to select it by default.

Example:

```
<!DOCTYPE html>
```

```
<html>
```

```
  <head>
```

```
    <title>Radio Box Control</title>
```

```
  </head>
```

```
  <body>
```

```
    <form>
```

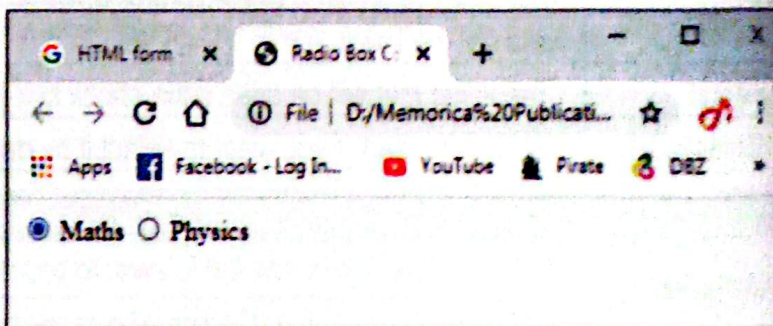
```
      <input type = "radio" name = "subject" value = "maths" checked> Maths
```

```
      <input type = "radio" name = "subject" value = "physics"> Physics
```

```
    </form>
```

```
  </body>
```

```
</html>
```



Select Box Control

A select box, also called a drop down box provides option to list down various options in the form of drop down list, from where a user can select one or more

options. `<select>` and `<option>` tags are used.

Attributes:

Attributes	Description
size	This can be used to present a scrolling list box.
name	Used to give a name to the control which is sent to the server to be recognized and get the value.
multiple	If set to "multiple" then allows a user to select multiple items from the menu.
value	The value that will be used if an option in the select box is selected.
selected	Specifies that this option should be the initially selected value when the page loads.

Example:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Select Box Control</title>
```

```
</head>
```

```
<body>
```

```
<form>
```

```
<select name = "dropdown" multiple size=5>
```

```
<option value = "Maths" selected>Maths</option>
```

```
<option value = "Physics">Physics</option>
```

```
<option value = "Computer">Computer</option>
```

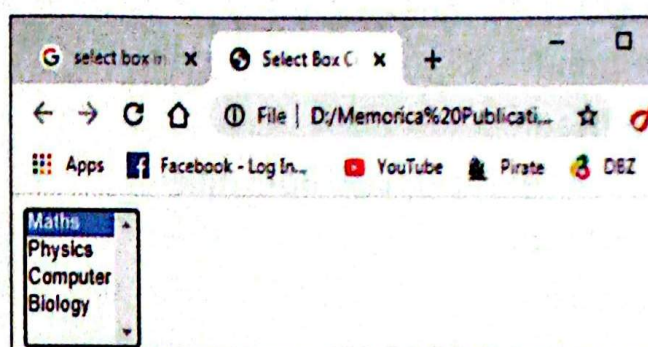
```
<option value = "Biology">Biology</option>
```

```
</select>
```

```
</form>
```

```
</body>
```

```
</html>
```



File Upload Box

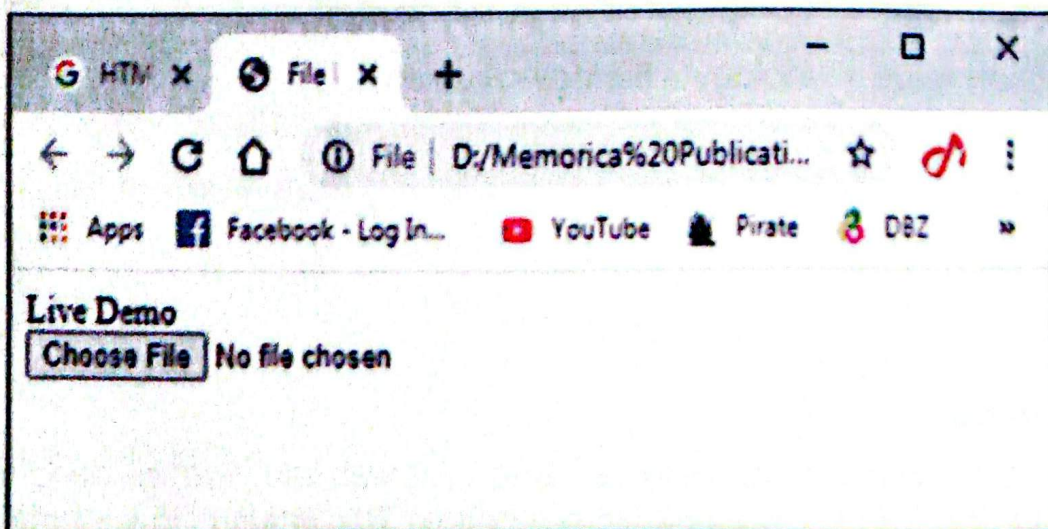
If you want to allow a user to upload a file to your web site, you will need to use a file upload box, also known as a file select box. This is also created using the `<input>` element but type attribute is set to file.

Attributes:

Attributes	Description
accept	Specifies the types of files that the server accepts. <input type="file" accept=".jpg,.png,.pdf">
multiple	Allows users to select multiple files at once <input type="file" multiple>
required:	Ensures that the user must select a file before submitting the form. <input type="file" required>

Example:

```
<!DOCTYPE html>
<html>
  <head>
    <title>File Upload Box</title>
  </head>
  <body>
    <p> Live demo</p>
    <form>
      <input type = "file" name = "fileupload" accept = "image/*" />
    </form>
  </body>
</html>
```



Button Controls

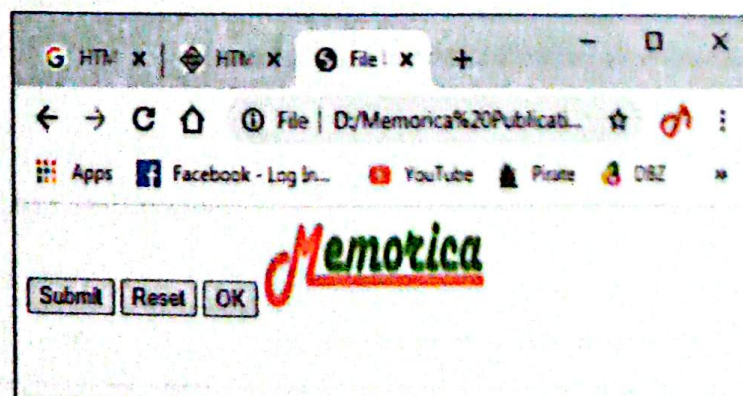
There are various ways in HTML to create clickable buttons. You can also create a clickable button using <input> tag by setting its type attribute to the button. The type attribute can take the following values.

Value	Description
submit	This creates a button that automatically submits a form.
reset	This creates a button that automatically resets form controls to their initial values.
button	This creates a button that is used to trigger a client-side script when the user clicks that button.
image	This creates a clickable button, but we can use an image as the background of the button.

Example:

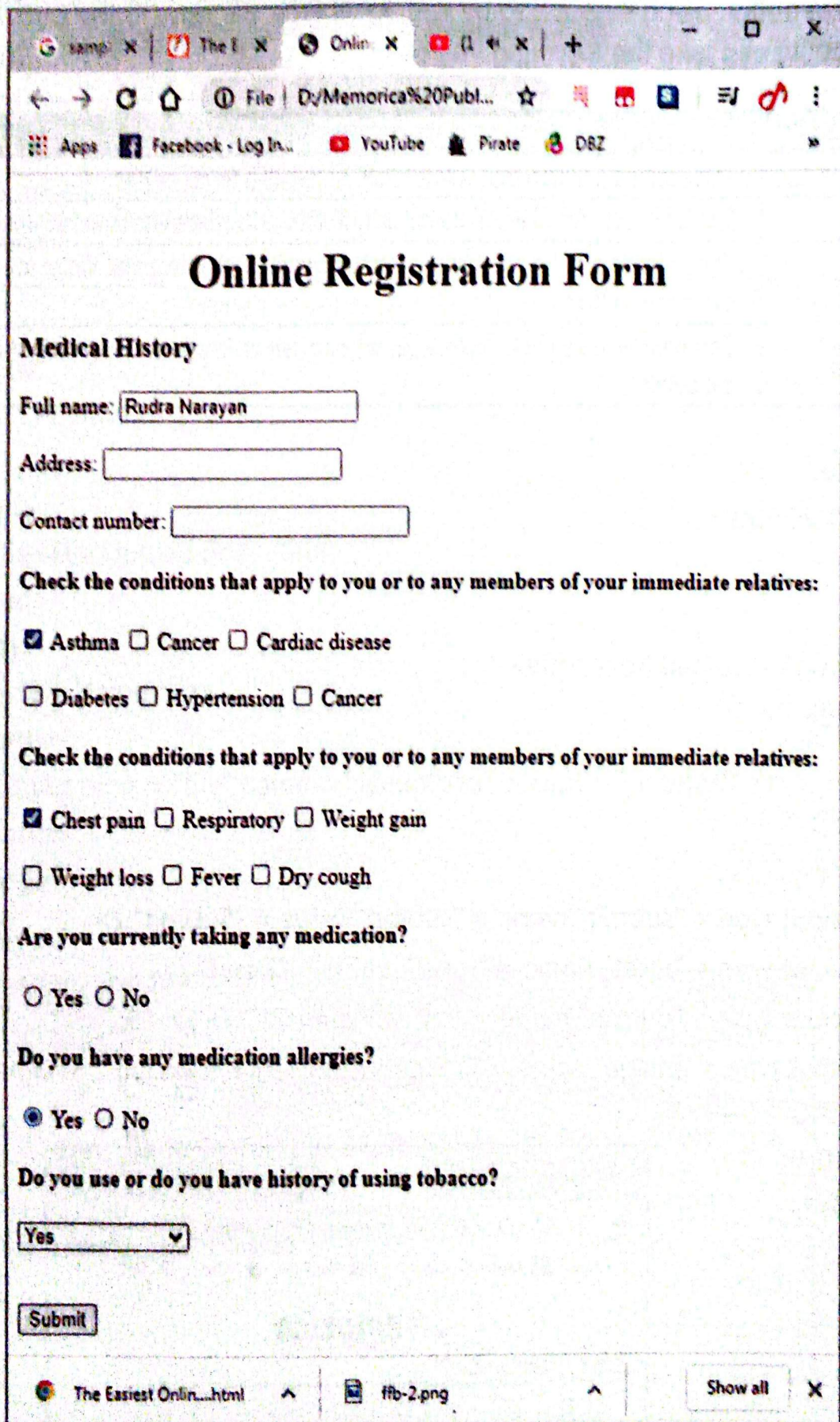
```
<!DOCTYPE html>
<html>
<head>
  <title>File Upload Box</title>
</head>

<body>
  <form>
    <input type = "submit" name = "submit" value = "Submit" />
    <input type = "reset" name = "reset" value = "Reset" />
    <input type = "button" name = "ok" value = "OK" />
    <input type = "image" name = "imagebutton" src="logo.jpg"; width="150"
      height="50"/>
  </form>
</body>
</html>
```



Lab Activity

1. Prepare the following online data entry form page using HTML.



The screenshot shows a web browser window with the following elements:

- Browser Tabs:** "smp", "The E", "Onlin", and "(1" (likely a download bar).
- Address Bar:** "File | D:/Memorica%20Publ...".
- Navigation Bar:** Includes icons for "Apps", "Facebook - Log In...", "YouTube", "Pirate", and "DBZ".
- Form Title:** "Online Registration Form".
- Section Header:** "Medical History".
- Form Fields:**
 - "Full name:" with a text box containing "Rudra Narayan".
 - "Address:" with an empty text box.
 - "Contact number:" with an empty text box.
- Checkboxes:**
 - Under "Check the conditions that apply to you or to any members of your immediate relatives:":
 - ☒ Asthma
 - ☐ Cancer
 - ☐ Cardiac disease
 - ☐ Diabetes
 - ☐ Hypertension
 - ☐ Cancer
 - Under another "Check the conditions that apply to you or to any members of your immediate relatives:":
 - ☒ Chest pain
 - ☐ Respiratory
 - ☐ Weight gain
 - ☐ Weight loss
 - ☐ Fever
 - ☐ Dry cough
- Questions:**
 - "Are you currently taking any medication?" with radio buttons for "Yes" and "No".
 - "Do you have any medication allergies?" with radio buttons for "Yes" (selected) and "No".
 - "Do you use or do you have history of using tobacco?" with a dropdown menu showing "Yes".
- Buttons:** A "Submit" button.
- Footer:** A taskbar at the bottom showing "The Easiest Onlin...html", "ffb-2.png", and a "Show all" button.

ii) HTML <div> tag

In HTML, the <div> tag is a block-level container that is used to arrange and group content. Although it does not apply any particular formatting or styling by default, it provides a flexible framework for section organization on a web page, which facilitates the application of styles and layout management with CSS. After learning CSS, using <div> is best practice.

Syntax:

```
<div>
    <!-- Content goes here -->
</div>
```

Example:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Div Tag Example</title>
<style>
    .container {
        background-color: #f2f2f2;
        padding: 20px;
    }
    .header, .content, .footer {
        margin-bottom: 20px;
    }
    .header {
        background-color: lightblue;
        padding: 10px;
    }
    .content {
        background-color: lightgreen;
        padding: 10px;
    }
    .footer {
```

```

        background-color: lightcoral;
        padding: 10px;
    }
</style>
</head>

<body>

    <div class="container">
        <div class="header">
            <h1>This is the Header Section</h1>
        </div>

        <div class="content">
            <p>This is the main content section.</p>
        </div>

        <div class="footer">
            <p>This is the footer section.</p>
        </div>
    </div>

</body>
</html>

```

